Argument-Structure-Based Feedback Prioritisation for English Argumentative Essays

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Group 04

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1. Problem and user

Problem

- Teachers must read every essay to find the weak arguments
- Writing detailed comments for a whole class takes hours
- Learners get a score, but not which part to fix

User and decision

- User: a teacher of English argumentative writing
- Decision 1: which essays to read first
- Decision 2: which segments need comments
- Decision 3: which essays the tool is unsure about
- Learners see a diagnosis only after the teacher approves it

2. Data

Dataset

- PERSUADE CORPUS 2.0 (Crossley et al., 2024, Assessing Writing)
- Route: public benchmark
- Argumentative essays by US students, grades 6–12
- Licence: CC BY-NC-SA 4.0 — academic, non-commercial
- In hand: official train set (15,594) and test set (10,402), no overlap

Size and fields

25,996	~238,000	13,121
essays · 15 prompts	labelled segments	independent essays

- Segment: argument role + effectiveness (Ineffective / Adequate / Effective)
- Essay: holistic score 1–6
- Prompt: used for prompt-based splitting
- Demographics (ELL, gender…): fairness checks only, never a model input

3. ML task

Three-step pipeline

- **Step 1 · Sentence role:** 8-class classification
  In: one sentence + its position in the essay
  Out: Lead / Position / Claim / Counterclaim / Rebuttal / Evidence / Concluding / none
- **Step 2 · Feedback priority:** 3-class classification
  In: segment text + role from Step 1 + position
  Out: Ineffective / Adequate / Effective
- **Step 3 · Essay level:** regression on 1–6
  In: ~45 rubric features (structure from Steps 1–2)
  Out: suggested level 1–6 + uncertainty
- **Subjective:** raters agree moderately on effectiveness (κ = 0.32), strongly on score (weighted κ = 0.75)
- **Imbalanced:** Adequate 79%, Ineffective 4.5%; 11× more Claims than Rebuttals

Two training examples, same prompt

	Weak	**Strong**
Essay	279E6B94B3C4 · 249 words	65B987E219BE · ~670 words
Sentence	*“ADD is a big reason too for people have no choice but to get distracted on things which cause crashes.”*	*“Cell phones are extremely distracting because of the large amount of focus operating one requires.”*
Step 1 · role	Claim	Claim
Step 2 · priority	**Ineffective**	**Effective**
Step 3 · level	**3**	**5**

4. Model plan

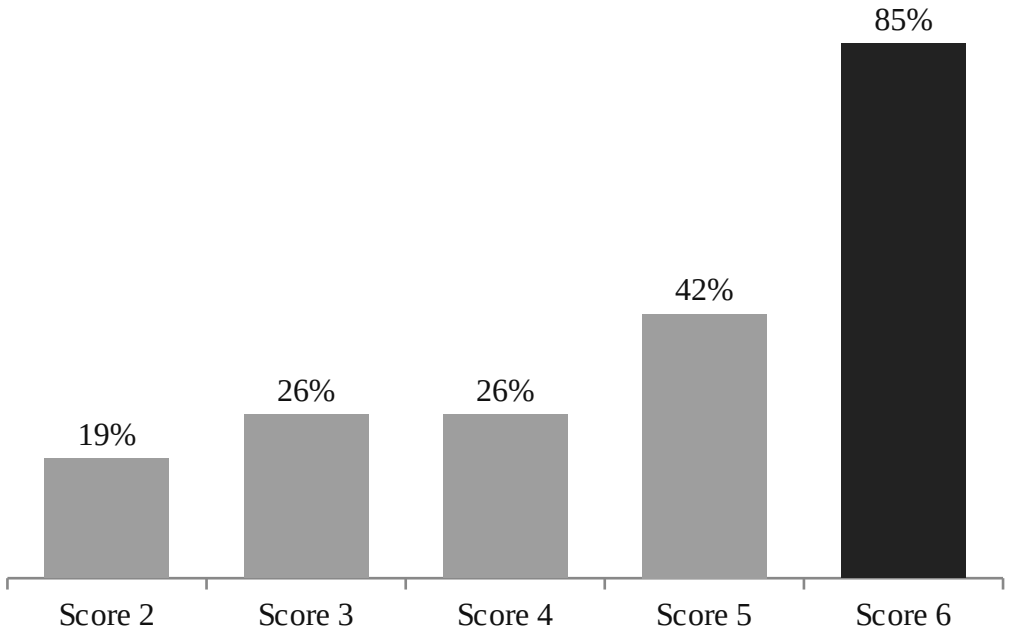
Logistic Regression → Steps 1, 2

- Strong and fast on sparse TF-IDF text
- Readable weights show which words flagged a segment
- Balanced class weights for Rebuttal, Ineffective

Random Forest, Gradient Boosting → Step 3

- Scores 4–6 need every criterion strong: a threshold, not a line
- Feature importance shows which criterion pulls the level down
- Uncertainty = spread of predictions across trees

Why trees: essays with a Counterclaim



Independent prompts, train set. Jump from 26% to 85% suits tree models.

5. Evaluation plan

Split and baselines

- Essay-level split: No essay overlap between sets.
- Prompt-level split: Hold out one prompt at a time.
- Training set: 15,594 essays; 10,669 remaining after excluding five prompts.
- Official test set (10,402) used once, at the end; 6,456 essays after excluding source-based prompts
- Steps 1–2 baseline: majority class; LR on position + length
- Step 3 baseline M0: word count only (r = 0.76 with score)
- Step 3 baseline M1: linguistic features (Yang et al., 2024)

Metrics

- Steps 1–2: macro-F1, recall on Ineffective
- Step 3: QWK, MAE
- Fairness by ELL status: CSD, MAED
- Ceiling: human agreement, weighted $\kappa = 0.75$

Why accuracy is not enough

Always predicting "Adequate" is 79% accurate yet finds no weak segment. Predicting 3 for a true 4 is not as bad as predicting 1.

6. Risks and responsible use

What could go wrong

- **Length shortcut:** word count r = 0.76 with score → test by duplicating paragraphs, stuffing connectives
- **Unfair to ELL learners:** mean 3.08 vs 3.73 → compare at the same true score
- **Noisy labels:** κ = 0.32 on effectiveness → a flag is a suggestion, not a verdict
- **Different population:** US grades 6–12 → no IELTS conversion, no high-stakes use

What stays human

- The teacher assigns the final grade
- The teacher can override any suggestion
- Learners see a diagnosis only after approval
- Every suggestion shows its confidence
- Demographics are never a model input

7. Contribution and next steps

Gap and research questions

- **Yang et al. (2024)**, *Unveiling the Tapestry of Automated Essay Scoring*, AAAI: accuracy and fairness on unseen prompts, but no argument structure
- **Wang et al. (2026)**, *ArguLens*, arXiv: argument structure, but gold labels only, no subgroup analysis
- **Chen, Wu & Zhang (2026)**, *Assessing Fairness in AI-Assisted Writing Scoring*, Assessing Writing: fairness conditioned on proficiency, but one prompt, no ELL status

- **RQ1:** how much does structure help on unseen prompts?
- **RQ2:** how much is lost with predicted instead of gold structure?
- **RQ3:** does structure increase or reduce bias against ELL learners?

Four versions of the Step-3 model

	Input	**Role**
M0	Word count only	Minimum baseline
M1	Linguistic features	Reproduces Yang et al.
M2	M1 + gold structure	Upper bound
M3	**M1 + predicted structure**	**Our system**

Same algorithm, same prompt-based split: only the inputs differ.

First step: build M0 and M1 with the prompt-based split